

Part B - Data Cleaning

Review the huggingface dataset and identify: (you may include but are not limited to)

1. Corrupted videos

```
[15]: corrupted_rows = []

for video in videos:
    cap = cv2.VideoCapture(str(video))
    opened = cap.isOpened()
    ok, frame = cap.read() if opened else (False, None)
    frame_count = int(cap.get(cv2.CAP_PROP_FRAME_COUNT)) if opened else 0
    fps = cap.get(cv2.CAP_PROP_FPS) if opened else 0
    duration = frame_count / fps if fps and fps > 0 else 0
    cap.release()

    if not opened or not ok or frame_count <= 0 or fps <= 0 or duration <= 0:
        corrupted_rows.append({
            "Issue Type": "Corrupted videos",
            "Affected Video": video.name,
            "Description": "Video cannot be opened/read properly, or has invalid frames, FPS, or duration.",
            "Suggested Data Cleaning method": "Remove the corrupted video or re-download it from the dataset."
        })

corrupted_df = pd.DataFrame(corrupted_rows)
print("Corrupted videos:", len(corrupted_df))
corrupted_df

Corrupted videos: 0
```

2. Incorrect filenames

```
[10]: actual_video_files = [v.name for v in videos]
metadata_files = set(df[file_col].dropna().astype(str))

missing_video_files = sorted(metadata_files - actual_video_files)
extra_video_files = sorted(actual_video_files - metadata_files)

incorrect_rows = []

for name in missing_video_files:
    incorrect_rows.append({
        "Issue Type": "Incorrect filenames",
        "Affected Video": name,
        "Description": "Filename appears in metadata but is missing from the videos folder.",
        "Suggested Data Cleaning method": "Check filename/path, rename if possible, otherwise remove the metadata record."
    })

for name in extra_video_files:
    incorrect_rows.append({
        "Issue Type": "Incorrect filenames",
        "Affected Video": name,
        "Description": "Video exists in folder but is not listed in metadata.",
        "Suggested Data Cleaning method": "Add metadata for this video or remove it from the cleaned dataset."
    })

incorrect_filenames_df = pd.DataFrame(incorrect_rows)
print("Incorrect filename issues:", len(incorrect_filenames_df))
incorrect_filenames_df

Incorrect filename issues: 0
```

3. Duplicate records

```
[17]: duplicate_rows = []

dup_meta = df[df.duplicated(subset=[file_col], keep=False)]

for _, row in dup_meta.iterrows():
    duplicate_rows.append({
        "Issue Type": "Duplicate records",
        "Affected Video": row[file_col],
        "Description": "Duplicate video filename found in metadata.",
        "Suggested Data Cleaning method": "Keep one valid record and remove duplicated metadata rows."
    })

def file_hash(path):
    h = hashlib.md5()
    with open(path, "rb") as f:
        for chunk in iter(lambda: f.read(1024 * 1024), b''):
            h.update(chunk)
    return h.hexdigest()

hash_df = pd.DataFrame({
    "file_name": [v.name for v in videos],
    "hash": [file_hash(v) for v in videos]
})

dup_hash_df = hash_df[hash_df.duplicated(subset=[hash], keep=False)]

for _, row in dup_hash_df.iterrows():
    duplicate_rows.append({
        "Issue Type": "Duplicate records",
        "Affected Video": row[file_name],
        "Description": "Duplicate video content detected using file hash.",
        "Suggested Data Cleaning method": "Keep one copy and remove duplicated video content."
    })

duplicates_df = pd.DataFrame(duplicate_rows)
print("Duplicate record issues:", len(duplicates_df))
duplicates_df
```

Duplicate record issues: 20

	Issue Type	Affected Video	Description	Suggested Data Cleaning method
0	Duplicate records	vid_0006.mp4	Duplicate video content detected using file hash.	Keep one copy and remove duplicated video cont...
1	Duplicate records	vid_0013.mp4	Duplicate video content detected using file hash.	Keep one copy and remove duplicated video cont...

4. Missing transcripts

```
[18]: missing_mask = (
    df[transcript_col].isna()
    | (df[transcript_col].astype(str).str.strip() == "")
    | (df[transcript_col].astype(str).str.lower().str.strip().isin(["nan", "none", "null"]))
)

missing_transcripts_df = df[missing_mask].copy()

missing_transcripts_report = pd.DataFrame({
    "Issue Type": "Missing transcripts",
    "Affected Video": missing_transcripts_df[file_col],
    "Description": "Transcript field is missing, empty, or invalid.",
    "Suggested Data Cleaning method": "Regenerate transcript using speech-to-text or remove the record if transcript is required."
})

print("Missing transcripts:", len(missing_transcripts_report))
missing_transcripts_report
```

Missing transcripts: 13

	Issue Type	Affected Video	Description	Suggested Data Cleaning method
3	Missing transcripts	vid_0004.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
383	Missing transcripts	vid_0384.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
534	Missing transcripts	vid_0535.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
545	Missing transcripts	vid_0546.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
721	Missing transcripts	vid_0722.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1003	Missing transcripts	vid_1004.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1175	Missing transcripts	vid_1176.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1335	Missing transcripts	vid_1336.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1376	Missing transcripts	vid_1377.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1409	Missing transcripts	vid_1410.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1445	Missing transcripts	vid_1446.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1656	Missing transcripts	vid_1657.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...
1966	Missing transcripts	vid_1967.mp4	Transcript field is missing, empty, or invalid.	Regenerate transcript using speech-to-text or ...